

Mohamed Charaf Eddine Deghboudj

AI & Data Science Engineering Student

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Profile

Third-year AI & Data Science engineering student at Algeria's National School of Artificial Intelligence (ENSIA), with hands-on experience building machine learning and optimization systems and leading national-scale technical education initiatives. Strong foundation in algorithms, statistical modeling, and full-stack development, combined with a track record of leading teams, training 20+ students directly, and representing Algeria in international technical competitions and delegations.

Education

Engineering Degree, Computer Science — AI & Data Science Specialization

2024 – Present

National School of Artificial Intelligence (ENSIA), Sidi Abdellah, Algiers

Five-year engineering program with an integrated preparatory cycle. Coursework: Data Structures & Algorithms, OOP, Theory of Computing & Formal Languages, Computer Architecture, Operating Systems, Databases (SQL/PL-SQL & Transactions), Introduction to AI, Mathematical Analysis, Statistics & Probability, Mathematical Logic, Concurrent Programming, Web Development, Linux, Electronics.

Baccalaureate Diploma, Mathematics Stream — Grade: Excellent (18.60/20)

2021 – 2024

Lycée Bali Chlali, Ksar El Abtal, Sétif

Excellence in Mathematics (20/20), Physics (19/20), and English (20/20).

Technical Skills

Programming Languages: C++, Python, Java, JavaScript, HTML/CSS

AI & Machine Learning: Optimization, Search Algorithms, Inference Logic, Machine Learning, Large Language Models (LLMs)

Systems & Infrastructure: Git, SQL/PL-SQL, Linux, Networks, Cloud Platforms, Bash Scripting

Web Development: Full-stack development — libraries, APIs, databases, deployment

Tools: LaTeX, Microsoft Office

Projects

Fuel Consumption Analytics & Flight Efficiency Platform (Machine Learning)

2026

AI Forge Program — World Learning / Algiers STEAM Makerspace

- Developed an AI-based analytics system to analyze real flight trajectories and fuel consumption.
- Applied predictive modeling, K-Means/DBSCAN clustering, and anomaly detection to identify inefficient flights and efficiency patterns across aircraft and routes.
- Designed a data-driven approach to support fuel-cost reduction and more sustainable airline operations.
- Awarded 1st place, Innovation Challenge (full program) — see Achievements.

Algiers in 24 Hours — Tourist Route Optimizer (Time-Constrained Orienteering Problem)

2026

Academic AI Project — ENSIA

- Modeled the Time-Constrained Orienteering Problem (TCOP) for tourist itinerary optimization.
- Implemented and benchmarked multiple metaheuristics: Genetic Algorithm, Tabu Search, Ant Colony Optimization (ACO), and GRASP.
- Built a Flask backend and a Next.js/Tailwind frontend, with CSV data processing and a database layer.

Online Booking System

2025 – 2026

Academic Project, Web Development — ENSIA

- Built a full-featured reservation management application with Booking.com API integration, secure authentication, a database layer, an admin dashboard, and payment management.

School Management System & Website

2022

Personal Project — Lycée Bali Chlali

- Built an internal management system and public-facing website for the school.

Experience

Community & Communications Manager / Coach & Trainer (Volunteer)

2025 – Present

Algerian Olympiads in Informatics, Mathematics & AI (AOI · AMO · AOAI)

- Coach students in competitive programming, AI, and problem-solving; lead training camps.
- Serve as Deputy Team Leader of Algeria's delegation at the North African Olympiad in Informatics (NAOI) 2026 and the Asia-Pacific Olympiad in Informatics (APIO) 2026.
- Draft official reports and coordinate administrative processes with the Ministry of Education; manage international participation logistics.
- Manage communication platforms, social media, brand identity, and content strategy for three national olympiads.

External Activities Manager

Aug. 2026 – Present

ENSIA Tech Community

- Manage external partnerships and community initiatives to grow the club's visibility, reach, and impact.
- Organize technology and awareness events for high school, middle school, and rural communities, focused on AI/technology ethics and digital skills.
- Develop collaborations with companies and organizations to create opportunities for members and strengthen the club's social impact.

Coach & Supervisor, Forsa Tech Hackathon

Dec. 2025

Algérie Télécom

- Mentored and supported participant teams during the national hackathon and datathon.
- Served as a member of the core organizing team.

Organizer, IndabaX 2024

Dec. 2024

Algeria's Leading AI Conference

- Helped organize Algeria's flagship national AI conference with the Skill&Tell Club.
- Handled logistics coordination, team management, and operational follow-up.

Cloud Platform Tester

Nov. 2025

ICOSNET

- Tested a new Algerian cloud service ahead of commercial launch, covering UX, technical, and business evaluation.
- Produced functional test reports and improvement recommendations.

Drone Piloting Trainee

Jun. 2025

CRTI — Centre for Industrial Technologies Research

- Completed theoretical training in meteorology, principles of flight, and aeronautical navigation.
- Completed a practical phase on a flight simulator and real outdoor drone piloting sessions.
- Obtained civil drone pilot license (certificate pending delivery).

Participant, AI Forge Programme

Mar. – Aug. 2026

World Learning Algeria & Boeing

- Completed theoretical training in Artificial Intelligence and Machine Learning.
- Explored AI applications in the aviation sector.
- Developed technical skills through real-world project work (see Projects).

Participant, UNESCO AI Ethics Programme 2026

Sep. 2026 – Present

LG AI Research

- Completed intensive training on AI impact, ethics, and governance.

Participant, GCI World AI Programme 2026

Sep. 2026 – Present

Matsuo Laboratory & University of Tokyo

- Completed intensive theoretical training in Artificial Intelligence and Machine Learning.

Founder & President, Computer Science Club

2022 – 2024

Lycée Bali Chlali

- Founded and led the club; trained 20+ students in programming, web development, and data structures.
- Organized online and on-site conferences with industry experts.
- Established partnerships with Huawei ICT Academy and MBI Algeria Campus; collaborated with Kings Academy on intensive on-site training sessions.

Active Member, University Scientific Clubs

2024 – Present

ETC Club & Skill&Tell Club – ENSIA

- External relations, Skill&Tell Club (2024).
- Events cell, ETC Club (2024); leadership role within the executive direction (2025).
- Student tutoring and mentoring; active participation in tech events.

Achievements & Distinctions

- **1st Place, Innovation Challenge** — AI Forge Program (World Learning & Boeing): developed an airline fuel-consumption anomaly detection and optimization system powered by AI/ML.
- **1st Place (National), Arab Code Week** — Smart City AI project; represented Algeria at the Arab-level round.
- **1st Place (National), Scientific Writing Competition** (Sonatrach & Ministry of National Education) — topic: digitalization and IT.
- **Deputy Team Leader**, Algerian delegation at NAOI 2026 (North Africa) and APIO 2026 (Asia-Pacific) Olympiads in Informatics.
- **Invited Mentor (Volunteer)**, AIESEC — Nasser Center for Science and Technology, Bahrain (Skill-Up Programme, 2026).
- **Volunteer**, UN Volunteers (UNV) and Algerian Red Crescent.

Certifications

- Mathematics Fundamentals Course — Huawei ICT Academy
- 21st Century Skills Training — World Learning Algeria (US Embassy, MBI Algeria Campus)
- AI Forge (Machine Learning) — World Learning Algeria

Languages

Arabic (Native) • English (C1 – Advanced) • French (C1 – Advanced)

Interests

Aviation & aeronautical systems • Applied AI & machine learning • Competitive programming • Hackathons & tech conferences